CHAPTER 4

SYSTEM IMPLEMENTATION, TESTING, AND RESULTS

This chapter describes the implementation of the Attorney Directory and Case File Archive System (PACFAS), the testing performed, the results obtained, and a discussion of the findings.

4.1 System Implementation

PACFAS was implemented as a web-based application using Node.js and Express.js for the backend, MySQL for relational storage, and HTML5/CSS3/Vanilla JavaScript for the frontend. The following subsections describe the implemented modules.

4.1.1 Login and Authentication Module

The login module authenticates users using registered credentials. Passwords are hashed with bcrypt, and sessions are managed through JSON Web Tokens (JWT). Role-based access control restricts features according to the user's role (Admin, Attorney, Staff, or User).

Figure 4.1. Login Interface - Insert a screenshot of the actual login page here.

4.1.2 Dashboard Module

The dashboard presents an overview of case statistics, including active, pending, and total document counts. It provides navigation to case management, clients, hearings, and (for administrators) attorney management. Search and filtering by status, category, and priority are supported.

Figure 4.2. Administrator Dashboard - Insert a screenshot of the actual dashboard here.

4.1.3 Case Management Module

This module allows authorized staff to create, view, search, and filter case files. Each case is assigned a category (Civil, Criminal, Corporate/Commercial, Family, Labor, Property, Tax, Intellectual Property, Immigration, Administrative, Environmental), a status (Active, Pending, Closed, Archived), and a priority (Low, Medium, High, Urgent). Case numbers are auto-generated in the format TYPE-YEAR-XXXX based on the selected category.

4.1.4 Document Archiving and OCR Module

Authorized users upload digital documents (PDF, PNG, JPEG, TIFF; maximum 25 MB). The system stores the file, creates a version record, and runs optical character recognition (OCR) in the background. Text-based PDFs are parsed directly; scanned (image-only) PDFs are rasterized and OCR'd page by page. Extracted text is stored and made searchable.

4.1.5 Hearing Schedule Management Module

This module allows attorneys and staff to schedule, update, and view court hearings linked to cases, including date, time, venue, courtroom, hearing type, and assigned judge. The system prevents scheduling conflicts by blocking bookings when an assigned attorney already has a hearing within one hour of the requested time (admin override available).

4.1.6 Attorney Directory Module

A public-facing directory lists verified attorneys with their specialization, credentials, and contact details. Visitors can view attorney profiles and public (non-confidential) case histories. Administrators can create, update, and deactivate attorney records.

4.1.7 Client Management Module

Authorized staff create and manage client profiles (name, birth date, contact, address) and link clients to cases with a party role (Plaintiff, Defendant, Petitioner).

4.1.8 Case Assignment Module

Staff assign primary and co-counsel attorneys to cases. Reassignments create new assignment records while preserving historical assignments. Inactive attorneys cannot be assigned to cases.

4.1.9 Admin Module

Administrators manage user accounts, attorney records, and role-based access. They may override scheduling conflicts and view the audit trail.

4.2 System Testing

System testing evaluates whether PACFAS functions according to its specified requirements. Testing covers authentication, case management, document upload and OCR, hearing scheduling, conflict detection, client and assignment management, and the public directory.

Test Case	Expected Result	Status
Login with valid credentials	User accesses the dashboard	Passed
Login with invalid credentials	Access is denied	Passed
Create a case	Case record is saved and case number generated	Passed
Upload a PDF document	Document stored and OCR text extracted	Passed
Upload an image document	OCR text extracted	Passed
Search document text	Matching documents returned	Passed
Schedule a hearing	Hearing record is saved	Passed
Overlapping hearing (same attorney, within 1 hour)	System detects the conflict	Passed
Admin override of conflict	Hearing is saved	Passed
Assign attorney to case	Assignment record created	Passed
Link client to case	Client-case link created	Passed
View attorney directory (public)	Attorneys listed	Passed
View public case summary	Non-confidential cases displayed	Passed
Create attorney (admin)	Attorney and user account created	Passed
Non-admin create attorney	Access denied (403)	Passed

Table 4.1. Functional Testing Results

Note: Replace sample statuses with actual observed outcomes if they differ. Remove any test case for a feature not implemented.

4.3 Results

The tested modules performed according to their expected behavior. Authentication, case management, document upload with OCR, hearing scheduling with conflict detection, client and assignment management, and the public directory all functioned as specified. Defects identified during testing (such as missing

dependencies and schema mismatches) were corrected prior to finalization.

4.4 Discussion

PACFAS provides a centralized way to manage case files, documents, hearings, clients, and attorney assignments. OCR-based text extraction and search reduce manual document retrieval. Conflict validation helps prevent overlapping hearing schedules. Centralized records make updates and retrieval more manageable than paper-based processes.

The system's effectiveness depends on the accuracy of entered data, the quality of scanned documents, and the results of testing under realistic conditions. Claims about efficiency or user satisfaction should be supported by measured data or user evaluation.

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATION

5.1 Summary

The study focused on the development of the Attorney Directory and Case File Archive System (PACFAS) to digitalize and centralize legal records for a public attorney office. It addresses challenges of physical document storage, slow file retrieval, and difficulty locating qualified attorneys.

The system includes modules for user authentication and role-based access, case management, document archiving with OCR, hearing scheduling with conflict prevention, attorney directory, client management, case assignment, and administration. It was implemented using Node.js/Express.js, MySQL, and web technologies, and tested through functional testing.

5.2 Conclusion

Based on the implementation and testing results, the tested functions support the management of case files, documents, hearings, clients, and attorney assignments. The system organizes legal records and makes them easier to retrieve and update. OCR-based search improves document retrieval, and conflict validation helps prevent scheduling overlaps.

The system may help organize scheduling and case information and make records easier to retrieve and update. Claims about improved efficiency, accuracy, or user satisfaction should be supported by measured results or documented user evaluation. The conclusion addresses each specific objective stated in Chapter 1.

5.3 Recommendation

1. For the Law Office Administration. Consider using the system to centralize case archiving and attorney directory management, provided it meets institutional requirements and passes necessary testing.

2. For Attorneys and Staff. Keep case data, client profiles, hearing schedules, and document records accurate before and after system use.

3. For Clients and the Public. Use the attorney directory and public case lookup to find qualified attorneys and view non-confidential case summaries.

4. For Future Researchers. Consider adding cloud storage and backup, email notifications, schedule optimization, and integration with existing court information systems.

5. For System Developers. Strengthen access controls, provide regular database backups, improve usability, and test a wider range of constraints and edge cases.

6. For Future System Enhancement. Consider support for multiple offices, advanced analytics, mobile access, and more complex scheduling and document-management rules.